Week 8 complete solutions

Problem-sheet solutions

Each solution states the intended method, relevant conditions, complete reasoning, mark allocation and specific common mistakes. Award method credit after a purely arithmetic slip when the inferential setup remains valid.

Question 1 — inferential objects and confidence

`W08-Q01-PA (S7)` **Method and conditions.** Separate the fixed population target from the sample function and then from its realised numerical value.

The population parameter is p, the proportion in the population. The estimand is also p: it is the population quantity the analysis targets. The estimator is

$$\hat{p} = \frac{X}{120},$$

which is a random variable because the count X varies across possible random samples. After observing $x = 84$, the estimate is

$$\hat{p}(\mathbf{x}) = \frac{84}{120} = 0.7.$$

This realised number is fixed for the observed data.

Final result. Parameter and estimand: p; estimator: $\hat{p} = X/120$; estimate: $\hat{p}(\mathbf{x}) = 0.7$.

Marks (5). Parameter, 1; estimand, 1; estimator, 1; numerical estimate, 1; random-before/fixed-after explanation, 1.

Common mistakes. Calling 0.7 the estimator; treating the fixed population parameter as random under the frequentist model; failing to distinguish the random count X from the observed count x.

`W08-Q01-PB (S8)` **Method and conditions.** Interpret confidence as long-run coverage of a random interval procedure under repeated samples satisfying the method's conditions.

Before sampling, the endpoints $L(\mathbf{X})$ and $U(\mathbf{X})$ are random because they are functions of the random sample. A 95% method is calibrated so that

$$\mathbb{P}_p(L(\mathbf{X}) \leq p \leq U(\mathbf{X})) = 0.95$$

exactly or approximately, according to the method and its conditions.

After the data are observed, the endpoints are realised numbers and the population parameter p is fixed. In the frequentist interpretation, the realised interval either covers p or it does not. The construction does not assign a 95% probability to that fixed proposition.

Final result and interpretation. Statement 1 is not the frequentist interpretation. Statement 2 is valid: approximately 95% of intervals from repeated use of the valid method cover the fixed p.

Marks (5). First statement assessed correctly, 1; repeated-sampling coverage stated, 2; random endpoints identified, 1; fixed realised interval/parameter distinction, 1.

Common mistakes. Assigning posterior probability to a fixed parameter; saying 95% of population observations lie inside the interval; omitting the conditions under which the method has its stated coverage.

Question 2 — a normal interval with known variance

`W08-Q02-PA (S8)` **Method and conditions.** The sample is iid from a normal population with known variance 36, so

$$\frac{\bar{X}-\mu}{6/\sqrt{36}} \sim \mathcal{N}(0,1)$$

exactly.

The standard error is

$$\frac{\sigma}{\sqrt{n}} = \frac{6}{6} = 1.$$

Thus the realised 95% confidence interval is

$$\begin{aligned}\left[\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right] &= [74 - 1.96,\ 74 + 1.96] \\ &= [72.04,\ 75.96].\end{aligned}$$

Final result and interpretation. The 95% interval is $[72.04, 75.96]$. Over repeated iid samples from the stated normal model, 95% of intervals constructed by this method cover μ. Normality and known variance make the pivot exact rather than a large-sample approximation.

Marks (6). Correct conditions and pivot, 1; standard error, 1; interval formula, 1; endpoints, 2; valid interpretation, 1.

Common mistakes. Treating 36 as the standard deviation; dividing by n rather than $\sqrt{n}$; using Student's t despite the known variance; assigning 95% probability to the fixed μ after observing the interval.

Question 3 — a small-sample Student's t interval

`W08-Q03-PA (S8)` **Method and conditions.** Compute the observed sample statistics, then use the one-sample t pivot. Independence and normality are material because the sample is small and the population variance is unknown.

The sample mean is

$$\bar{x} = \frac{8 + 10 + 9 + 13 + 10}{5} = 10.$$

The deviations from 10 are $-2, 0, -1, 3, 0$, whose squared values sum to 14. Therefore,

$$s^2 = \frac{14}{5-1} = 3.5.$$

The estimated standard error is

$$\frac{s}{\sqrt{n}} = \sqrt{\frac{s^2}{n}} = \sqrt{\frac{3.5}{5}} = \sqrt{0.7}.$$

Under the stated model,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{5}} \sim t_4.$$

The realised interval is

$$\begin{aligned}\left[10 - 2.776\sqrt{0.7}, 10 + 2.776\sqrt{0.7}\right] &\approx [10 - 2.323,\ 10 + 2.323] \\ &= [7.677,\ 12.323].\end{aligned}$$

Final result. The 95% Student's t interval is approximately $[7.677, 12.323]$, using 4 degrees of freedom.

Marks (7). Mean, 1; squared-deviation sum and sample variance, 2; reference distribution and degrees of freedom, 1; standard error, 1; interval endpoints, 1; conditions, 1.

Common mistakes. Dividing the squared deviations by n; using a standard-normal critical value; using 5 rather than 4 degrees of freedom; omitting the normal-population condition for the exact small-sample method.

Question 4 — a proportion interval and planned precision

`W08-Q04-PA (S8)` **Method and conditions.** Use the large-sample normal interval for a proportion after checking the observed success and failure counts.

The estimate is

$$\hat{p} = \frac{120}{200} = 0.6.$$

The counts are $200(0.6) = 120$ successes and $200(0.4) = 80$ failures, both large enough for the stated normal approximation. The estimated standard error is

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.6(0.4)}{200}} \approx 0.0346.$$

The 90% interval is

$$[0.6 - 1.645(0.0346), 0.6 + 1.645(0.0346)] \approx [0.543,\ 0.657].$$

Final result and interpretation. The approximate 90% interval is $[0.543, 0.657]$. Under independent random sampling and repeated use of this large-sample method, approximately 90% of the intervals cover the population proportion p.

Marks (6). Estimate, 1; count check and independence condition, 1; standard error, 1; interval setup, 1; endpoints, 1; valid interpretation, 1.

Common mistakes. Using 120 as the sample size; failing to check both successes and failures; using $p = 0.5$ in the observed interval; claiming that 90% of applicants have probabilities inside the interval.

`W08-Q04-PB (S8)` **Method and conditions.** Use the maximum possible Bernoulli variance $p(1-p) = 1/4$ to guarantee the margin bound without a prior value of p.

The conservative margin is

$$1.96\sqrt{\frac{1/4}{n}}.$$

Requiring it to be no greater than 0.04 gives

$$1.96\sqrt{\frac{1}{4n}} \leq 0.04,$$
$$n \geq \frac{1.96^2(1/4)}{0.04^2}$$
$$= 600.25.$$

Because n must be an integer and the margin must not exceed the target, round upwards.

Final result and interpretation. The smallest planned sample size is $n = 601$. The choice $p(1-p) \leq 1/4$ is conservative for every $p \in [0,1]$, so it does not require a pilot estimate of the unknown proportion.

Marks (5). Conservative variance bound, 1; margin inequality, 1; algebraic lower bound, 1; upward rounding, 1; explanation of conservatism, 1.

Common mistakes. Using 0.5 rather than $0.5(1-0.5)$ as the variance factor; rounding 600.25 down; treating 0.04 as the full interval width; claiming that the calculation guarantees coverage without random-sampling and approximation conditions.

Question 5 — a one-sided z test

`W08-Q05-PA (S9)` **Method and conditions.** The concern determines a lower-tailed test. Under the stated iid normal model with known variance,

$$H_0 : \mu = 50 \qquad \text{and} \qquad H_1 : \mu < 50.$$

The standard error under the model is

$$\frac{\sigma}{\sqrt{n}} = \frac{4}{4} = 1.$$

The observed test statistic is

$$z_{\text{obs}} = \frac{48.5 - 50}{1} = -1.5.$$

For the lower-tailed alternative, the p-value is

$$\mathbb{P}_{H_0}(Z \leq -1.5) = 0.0668.$$

Because $0.0668 > 0.05$, do not reject H_0. Equivalently, $-1.5 > -1.645$, so the statistic is not in the lower-tail rejection region.

Final result. $z_{\text{obs}} = -1.5$, the p-value is 0.0668, and H_0 is not rejected at the 5% level.

Marks (6). Hypotheses and lower-tail direction, 1; standard error, 1; statistic, 1; p-value setup and value, 2; decision, 1.

Common mistakes. Using a two-sided alternative despite the stated concern; taking the upper tail; treating 16 as the standard deviation; rejecting because the observed mean is numerically below 50.

`W08-Q05-PB (S9)` **Method and conditions.** Interpret the p-value using the null model and the tail chosen before observing the data.

If $H_0 : \mu = 50$ were true, the p-value 0.0668 is the probability of obtaining a sample-mean z statistic no greater than -1.5, which is at least as incompatible with H_0 in the specified lower-tail direction as the observed result.

Non-rejection means that the result does not cross the chosen 5% evidence threshold. It does not prove equality $\mu = 50$, and the frequentist test does not calculate $\mathbb{P}(H_0 \mid \text{data})$.

Final result and interpretation. The data provide insufficient evidence at level 0.05 for the claim $\mu < 50$; they do not establish that the null is true.

Marks (4). Null-model probability, 1; at-least-as-incompatible lower-tail event, 1; non-rejection interpreted correctly, 1; probability-of-null error corrected, 1.

Common mistakes. Calling 0.0668 the probability that H_0 is true; interpreting non-rejection as proof of equality; forgetting that the p-value uses the pre-specified lower tail.

Question 6 — a two-sided t test and its interval

`W08-Q06-PA (S9)` **Method and conditions.** The observations are independent and normal, and the population variance is unknown. Under H_0,

$$T = \frac{\bar{X} - 50}{S/\sqrt{10}} \sim t_9.$$

The observed statistic is

$$t_{\text{obs}} = \frac{52 - 50}{3/\sqrt{10}} \approx 2.108.$$

Because the alternative is two-sided, the p-value counts both tails:

$$p\text{-value} = 2\mathbb{P}(T_9 \geq 2.108) \approx 2(0.032) = 0.064.$$

Since $0.064 > 0.05$, do not reject H_0. The critical-value check agrees because $|2.108| < 2.262$.

Final result. $t_{\text{obs}} \approx 2.108$, with 9 degrees of freedom; the two-sided p-value is approximately 0.064, so H_0 is not rejected at the 5% level.

Marks (7). Conditions and t reference distribution, 1; degrees of freedom, 1; statistic setup and value, 2; two-sided p-value, 2; decision, 1.

Common mistakes. Using a z statistic with unknown variance and small sample; using 10 degrees of freedom; failing to double the one-tail probability; rejecting because the statistic is positive.

`W08-Q06-PB (S8, S9)` **Method and conditions.** Use the same t pivot, estimated standard error, degrees of freedom and two-sided tail allocation as in Part A.

The estimated standard error is

$$\frac{s}{\sqrt{n}} = \frac{3}{\sqrt{10}}.$$

The realised 95% interval is

$$\begin{aligned}\left[52 - 2.262\frac{3}{\sqrt{10}}, 52 + 2.262\frac{3}{\sqrt{10}}\right] &\approx [52 - 2.146,\ 52 + 2.146] \\ &= [49.854,\ 54.146].\end{aligned}$$

The null value 50 lies inside the interval, so the matched two-sided test does not reject it at $\alpha = 0.05$. This duality requires the same two-sided level, t pivot, standard-error estimate, degrees of freedom and assumptions for both procedures.

Final result and interpretation. The 95% interval is approximately $[49.854, 54.146]$, which contains 50 and therefore agrees with the non-rejection decision.

Marks (5). Interval formula, 1; endpoints, 2; containment and matched decision, 1; conditions for duality, 1.

Common mistakes. Using a normal critical value; comparing the sample mean rather than the null value with the interval; claiming duality for unmatched one-sided and two-sided procedures; interpreting containment as proof that $\mu = 50$.

Question 7 — a large-sample proportion test

`W08-Q07-PA (S9)` **Method and conditions.** Under the null, $X \sim \text{Binomial}(100, 0.5)$. The null expected success and failure counts are both 50, so the normal approximation is appropriate under the stated independent Bernoulli model.

The observed proportion is $\hat{p} = 58/100 = 0.58$. The null standard error is

$$\sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{0.5(0.5)}{100}} = 0.05.$$

Thus

$$z_{\text{obs}} = \frac{0.58 - 0.5}{0.05} = 1.6.$$

The two-sided p-value is

$$2\mathbb{P}(Z \geq 1.6) = 2(1 - 0.9452) = 0.1096.$$

Since $0.1096 > 0.05$, do not reject H_0. Equivalently, $|1.6| < 1.96$.

Final result. The observed z statistic is 1.6, the two-sided p-value is 0.1096, and $H_0 : p = 0.5$ is not rejected at the 5% level.

Marks (6). Null count check and conditions, 1; null standard error, 1; statistic, 1; two-sided p-value, 2; decision, 1.

Common mistakes. Using the observed proportion in the null standard error; checking observed rather than null success and failure counts; failing to double the upper-tail probability; interpreting non-rejection as proof that $p = 0.5$.

Question 8 — Type I error, Type II error and power

`W08-Q08-PA (S9)` **Method and conditions.** The sample mean has the exact distribution

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{16}{16}\right) = \mathcal{N}(\mu, 1).$$

The Type I error probability is the rejection probability at the null value:

$$\begin{aligned}\alpha &= \mathbb{P}_{\mu=50}(\bar{X} > 52)\\ &= \mathbb{P}(Z > 2)\\ &= 1 - 0.9772\\ &= 0.0228.\end{aligned}$$

At the specified alternative $\mu = 53$, a Type II error is non-rejection:

$$\begin{aligned}\beta(53) &= \mathbb{P}_{\mu=53}(\bar{X} \le 52)\\ &= \mathbb{P}(Z \le -1)\\ &= 0.1587.\end{aligned}$$

Therefore the power at $\mu = 53$ is

$$1 - \beta(53) = 0.8413.$$

Final result. $\alpha = 0.0228$, $\beta(53) = 0.1587$, and the power at $\mu = 53$ is 0.8413.

Marks (6). Sampling standard deviation, 1; Type I event and probability, 1; Type II event and probability, 2; power, 1; alternative value identified, 1.

Common mistakes. Assuming the rule automatically has $\alpha = 0.05$; calculating both errors under the null; defining Type II error as rejection at $\mu = 53$; reporting power without naming the alternative value.

`W08-Q08-PB (S9)` **Method and conditions.** First calibrate the boundary to have rejection probability 0.05 under $\mu = 50$. Then impose the power requirement at the specified alternative.

For a sample of size n,

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{16}{n}\right).$$

The level condition is satisfied by

$$c_n = 50 + z_{0.95}\frac{4}{\sqrt{n}} = 50 + 1.645\frac{4}{\sqrt{n}}.$$

At $\mu = 52$, power at least 0.90 is equivalent to $\beta(52) \le 0.10$. Hence,

$$\begin{aligned}\mathbb{P}_{\mu=52}(\bar{X} \le c_n) &\le 0.10,\\ \frac{c_n - 52}{4/\sqrt{n}} &\le z_{0.10} = -1.282.\end{aligned}$$

Substituting c_n gives

$$\begin{aligned}\frac{50 - 52}{4/\sqrt{n}} + 1.645 &\le -1.282,\\ -\frac{\sqrt{n}}{2} &\le -2.927,\\ \sqrt{n} &\ge 5.854,\\ n &\ge 34.269.\end{aligned}$$

Thus the smallest integer candidate is $n = 35$.

For $n = 34$, the standardised boundary under $\mu = 52$ is approximately -1.270, which is greater than -1.282. Its Type II error therefore exceeds 0.10, so its power is below 0.90. For $n = 35$, the standardised boundary is approximately $-1.313 < -1.282$, so the power exceeds 0.90.

Final result. The smallest sample size is $n = 35$, with rejection boundary

$$c_{35} = 50 + 1.645\frac{4}{\sqrt{35}} \approx 51.112.$$

Marks (10). Sampling distribution, 1; level-calibrated boundary, 2; Type II or power inequality, 2; standardisation at $\mu = 52$, 1; sample-size algebra, 2; upward rounding and adjacent-integer verification, 2.

Common mistakes. Using a two-sided critical value for the upper-tailed test; calibrating the boundary under $\mu = 52$ rather than the null; setting the Type II error equal to 0.90; rounding the sample-size lower bound down.

Question 9 — test–interval duality and non-rejection

`W08-Q09-PA (S8, S9)` **Method and conditions.** For a two-sided level-0.05 z test and the 95% z interval obtained from the same pivot and standard error, reject a candidate null value exactly when it lies outside the realised interval.

The value 100 lies below the interval $[102.4, 107.6]$. Therefore reject $H_0 : \mu = 100$ against the two-sided alternative at $\alpha = 0.05$.

The value 105 lies inside the interval. Therefore do not reject $H_0 : \mu = 105$ against the two-sided alternative at $\alpha = 0.05$.

The duality holds because the interval and tests use the same normal model, standard error, two-sided tail allocation and significance/confidence level. Non-rejection of 105 means only that 105 is not excluded by this procedure at the 5% level. It does not assign 95% probability to the point hypothesis $\mu = 105$.

Final result. Reject the null value 100; do not reject the null value 105. Neither decision assigns a probability to the truth of a fixed null hypothesis.

Marks (6). Decision for 100, 1; decision for 105, 1; interval-containment duality, 2; matching-procedure condition, 1; probability claim corrected, 1.

Common mistakes. Reversing the containment rule; treating non-rejection as acceptance or proof; claiming duality when the test and interval use different tails or standard errors; assigning posterior probability to 105.

Question 10 — earlier Statistics retrieval

`W08-Q10-PA (S6)` **Method and conditions.** Use iid sampling to obtain the first two moments of the sample mean, then apply the central limit theorem because the population distribution is not specified as normal.

$$\mathbb{E}[\bar{X}] = 12, \qquad \mathrm{Var}(\bar{X}) = \frac{9}{36} = \frac{1}{4}.$$

Thus the standard deviation of $\bar{X}$ is $1/2$. The central limit theorem gives

$$\bar{X} \mathrel{\dot\sim} \mathcal{N}\left(12, \frac{1}{4}\right),$$

where $\dot\sim$ denotes an approximation. Therefore,

$$\begin{aligned}\mathbb{P}(11.5 \leq \bar{X} \leq 12.5) &\approx \mathbb{P}\left(\frac{11.5-12}{0.5} \leq Z \leq \frac{12.5-12}{0.5}\right)\\ &= \mathbb{P}(-1 \leq Z \leq 1)\\ &= 0.6826.\end{aligned}$$

Final result. $\mathbb{E}[\bar{X}] = 12$, $\mathrm{Var}(\bar{X}) = 1/4$, and the requested probability is approximately 0.6826. The calculation is approximate because only the first two population moments, not a normal population distribution, are specified.

Marks (4). Mean and variance, 1; standardisation, 1; probability, 1; CLT/approximation qualification, 1.

Common mistakes. Using variance 9 for the sample mean; dividing the standard deviation by n; claiming an exact normal distribution without a normal population assumption; omitting the approximation sign.

`W08-Q10-PB (S7)` **Method and conditions.** Apply the expectation and variance formulae for an iid sample mean, then use convergence of its variance.

Since $\mathbb{E}[X_i] = p$ and $\mathrm{Var}(X_i) = p(1-p)$,

$$\mathbb{E}[\hat{p}] = \mathbb{E}[\bar{X}] = p.$$

Thus the bias is zero. Independence gives

$$\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

For every $\varepsilon > 0$, Chebyshev's inequality gives

$$\mathbb{P}\left(|\hat{p} - p| \geq \varepsilon\right) \leq \frac{p(1-p)}{n\varepsilon^2} \longrightarrow 0.$$

Hence $\hat{p}$ is consistent for the estimand p.

For the observed sample,

$$\hat{p}(\mathbf{x}) = \frac{7}{20} = 0.35.$$

Final result. The estimator is unbiased, has variance $p(1-p)/n$, is consistent for p, and gives the observed estimate 0.35.

Marks (4). Bias, 1; variance, 1; consistency argument, 1; observed estimate, 1.

Common mistakes. Calling 0.35 the estimator rather than the estimate; omitting the division by n in the variance; claiming consistency merely because the estimator is unbiased; treating the unknown parameter p as the observed sample proportion.

Question 11 — earlier Mathematics retrieval

`W08-Q11-PA (M10, M11)` **Method and conditions.** Use the displayed equality encoding and the course-wide maximisation convention $g(x, y) = y \geq 0$, with $\lambda \geq 0$. Solve the complementary-slackness cases, check each regime and use concavity for sufficiency.

The Kuhn–Tucker conditions are:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial x} &= 4 - 2x - \mu = 0, \\
\frac{\partial \mathcal{L}}{\partial y} &= 2 - \mu + \lambda = 0, \\
a - x - y &= 0, \\
y &\geq 0, \\
\lambda &\geq 0, \\
\lambda y &= 0.
\end{aligned}$$

For an interior solution, $y > 0$, so complementary slackness requires $\lambda = 0$. The y-stationarity condition gives $\mu = 2$, and the x-stationarity condition then gives $x = 1$. Equality feasibility gives $y = a - 1$. This is an interior solution exactly when $a > 1$.

For a corner solution, $y = 0$, so equality feasibility gives $x = a$. Stationarity gives

$$\mu = 4 - 2a \qquad \text{and} \qquad \lambda = \mu - 2 = 2 - 2a.$$

Dual feasibility requires $\lambda \geq 0$, which holds exactly when $0 < a \leq 1$. At $a = 1$, the optimum is the corner $(x, y) = (1, 0)$ with $\lambda = 0$.

For a global check, substitute $y = a - x$. Feasibility becomes $x \leq a$, and the objective becomes

$$\phi_a(x) = 2a + 2x - x^2.$$

Because $\phi_a''(x) = -2 < 0$, each feasible problem is strictly concave and the candidate in the applicable regime is its unique global maximiser.

The value function is therefore

$$V(a) = \begin{cases} 4a - a^2, & 0 < a \leq 1, \\ 2a + 1, & a > 1. \end{cases}$$

The equality is encoded as $h(x, y, a) = a - x - y = 0$. The envelope theorem gives

$$V'(a) = \left.\frac{\partial \mathcal{L}}{\partial a}\right|_{(x^*, y^*, \lambda^*, \mu^*)} = \mu^*(a) = \begin{cases} 4 - 2a, & 0 < a \leq 1, \\ 2, & a > 1. \end{cases}$$

Both expressions equal 2 at $a = 1$. Under this encoding, $\mu^*(a)$ is the marginal increase in the maximised objective from a small increase in the required total $x + y = a$, subject to remaining in the applicable regime.

Final result.

$$(x^*(a), y^*(a)) = \begin{cases} (a, 0), & 0 < a \leq 1, \\ (1, a - 1), & a > 1, \end{cases}$$

with $V(a)$ and $\mu^*(a) = V'(a)$ as displayed above.

Marks (8). Complete Kuhn–Tucker conditions, 2; interior solution and regime, 1; corner solution and dual-feasibility regime, 2; global sufficiency check, 1; value function, 1; envelope derivative and multiplier interpretation, 1.

Common mistakes. Using a negative multiplier with the course convention $y \geq 0$; omitting complementary slackness; failing to check both the interior feasibility and corner dual-feasibility regimes; interpreting μ without reference to the encoding $a - x - y = 0$.

Self-test solutions and error check

Solutions and error check

Solution to `W08-ST01-PA (S8)` Method and conditions. The independent sample comes from a normal population and the variance is unknown, so use Student's t with $n - 1 = 8$ degrees of freedom.

The estimated standard error is

$$\frac{s}{\sqrt{n}} = \frac{3}{3} = 1.$$

Thus the realised 95% interval is

$$[15 - 2.306,\ 15 + 2.306] = [12.694,\ 17.306].$$

In repeated iid samples from the stated normal model, 95% of intervals constructed by this method cover μ.

Marks (5). Method, condition and degrees of freedom, 2; standard error, 1; endpoints, 1; valid interpretation, 1.

Common mistakes. Using a z critical value with unknown variance; using 9 degrees of freedom; assigning 95% probability to the fixed μ.

Solution to `W08-ST02-PA (S8)` Method and conditions. The estimate is $\hat{p} = 40/100 = 0.4$. There are 40 successes and 60 failures, so the large-sample normal approximation is reasonable under independent random sampling.

The realised interval is

$$[0.4 - 1.96(0.0490), 0.4 + 1.96(0.0490)] \approx [0.304,\ 0.496].$$

In repeated independent samples, approximately 95% of intervals constructed by this large-sample method cover the fixed population proportion p. For the observed data, the endpoints are fixed; the interval does not assign a 95% probability to p.

Marks (5). Estimate, 1; two count checks and independence condition, 1; standard error or valid interval setup, 1; endpoints, 1; repeated-sampling interpretation, 1.

Common mistakes. Checking only the success count; using 40 as the sample size; using $p = 0.5$ in the observed interval; interpreting the interval as a range containing 95% of individual responses.

Solution to `W08-ST05-PA (S9)` Method and conditions. The observations are independent and normal, the population variance is unknown, and the alternative selects the upper tail. Under H_0,

$$T = \frac{\bar{X} - 20}{S/\sqrt{16}} \sim t_{15}.$$

The observed statistic is

$$t_{\text{obs}} = \frac{21.5 - 20}{4/4} = 1.5.$$

The upper-tail p-value is 0.077. Because $0.077 > 0.05$, do not reject H_0.

The p-value is the null probability of a t statistic at least as large as 1.5 in the stated upper-tail direction. It is not the probability that H_0 is true, and non-rejection does not prove that $\mu = 20$.

Marks (5). Student's t method, upper-tail direction and degrees of freedom, 1; standard error and statistic, 1; p-value, 1; decision, 1; interpretation, 1.

Common mistakes. Using a standard-normal reference distribution despite the unknown variance; doubling the one-sided p-value; using the lower tail; treating non-rejection as proof that H_0 is true.

Solution to `W08-ST04-PA (S9)` Method and conditions. The sample mean has distribution

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{4}{4}\right) = \mathcal{N}(\mu, 1).$$

At the null value,

$$\alpha = \mathbb{P}_{\mu=0}(\bar{X} > 1.645) = \mathbb{P}(Z > 1.645) = 0.05.$$

At $\mu = 2$, a Type II error is non-rejection:

$$\begin{aligned} \beta(2) &= \mathbb{P}_{\mu=2}(\bar{X} \leq 1.645) \\ &= \Phi(1.645 - 2) \\ &= \Phi(-0.355) \\ &= 0.3613. \end{aligned}$$

The power at $\mu = 2$ is

$$1 - \beta(2) = 0.6387.$$

Marks (5). Sampling distribution, 1; Type I probability, 1; Type II event and standardisation, 1; Type II value, 1; power, 1.

Common mistakes. Treating 4 as the standard deviation; computing Type II error under the null; using the rejection event for $\beta(2)$; reporting power without specifying $\mu = 2$.

Record the first cause of each error under `S8` or `S9`, then reattempt that part from a blank page before the Statistics mock.